

XTrace & XNysTrace

Optimal Stochastic Trace Estimation

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Introduction

Trace estimation is a randomized linear algebra technique for approximating the trace of a matrix without explicitly computing or storing the entire matrix. This makes it particularly useful for larger problems where constructing the full matrix would be computationally or memory intensive.

Trace estimation is carried out by performing only matrix-vector(matvec) products, allowing us to avoid these potential issues caused by the size of the matrix.

Girard-Hutch and Hutch++ Estimator

The earliest method of trace estimation was proposed by Girard, then later expanded on by Hutchinson. While this first method is both unbiased and exchangeable, it converges at the Monte-Carlo rate($\Theta(m^{-1})$) as we increase the number of samples m , making it rather inefficient.

Later methods implement variance reduction techniques to improve the rate of convergence. Hutch++ forms a low-rank approximation of the input matrix, making it so that we only need to compute the trace of the residual. This significantly reduces the variance of the estimator and improves convergence rate. As a tradeoff, however, the resulting estimator is no longer exchangeable.

The Exchangeability Principle

Exchangeability Principle

If the test vectors $\omega_1, \dots, \omega_k$ are exchangeable, then the minimum variance unbiased estimator for $Q(A)$ is always a symmetric function of $\omega_1, \dots, \omega_k$.

Essentially, the family $(\omega_1, \dots, \omega_k)$ has the same distribution as the permuted family of $\omega_{\sigma 1}, \dots, \omega_{\sigma k}$ for every permutation σ in the symmetric group S_k .

Key Takeaway: Exchangeability means that we do not care about the order in which the random vectors are sampled, guaranteeing that averaging over all permutations preserves unbiasedness and can only reduce variance.

What is XTrace?

The Issue with Hutch++

Hutch++ splits the test vectors into two groups:

- ① One half for low-rank approximation
 - ② The second half for calculating residual trace estimation
- Violates the exchangeability principle since the estimators must be symmetric over all test vectors.
 - However, if we were to symmetrize Hutch++, we would run into computational infeasibility due to combinatorial explosion of vector assignments.

The Solution is in XTrace

XTrace utilizes an unbalanced split:

- ① $m/2$ vectors used for low-rank approximation
 - ② Only 1 vector used for residual trace estimation
- By symmetrizing via the Leave-One-Out (LOO) method, every vector will serve in both sides of the split.
 - This results in the same $\mathcal{O}(m^2N)$ computational cost as Hutch++, while significantly lowering variance and satisfying the exchangeability principle.

XTrace Algorithm

Setup and Notation

- $\mathbf{A} \in \mathbb{R}^{N \times N}$: square matrix input
 - m : number of matrix vector products (even)
 - ω_i : i.i.d isotropic test vectors ($i = 1, \dots, m/2$)
 - $\Omega = [\omega_1 \cdots \omega_{m/2}]$: Test Matrix
- ❶ **LOO subspace.** For each $i = 1, \dots, m/2$, drop the i -th vector i :

$$\mathbf{Q}_{(i)} = \text{orth}(\mathbf{A}\Omega_{-i})$$

Each vector ω_i plays a symmetric role.

- ❷ **Basic estimator.** Each t_i splits into:

$$\hat{\text{tr}}_i = \underbrace{\text{tr}(\mathbf{Q}_{(i)}^* \mathbf{A} \mathbf{Q}_{(i)})}_{\text{low-rank part}} + \underbrace{\omega_i^* (\mathbf{I} - \mathbf{Q}_{(i)} \mathbf{Q}_{(i)}^*) \mathbf{A} (\mathbf{I} - \mathbf{Q}_{(i)} \mathbf{Q}_{(i)}^*) \omega_i}_{\text{residual part}}$$

- ❸ **XTrace estimator.**

$$\hat{\text{tr}}_X = \frac{1}{m/2} \sum_{i=1}^{m/2} \hat{\text{tr}}_i$$

Symmetric average over all LOO estimators satisfies exchangeability principle.

Efficient Implementation

Key Insight: avoids recomputing each LOO subspace from scratch every time.

1
Full QR Decomposition
 Compute $A\Omega = QR$ only once. This decomp. drives the following LOO updates.

2
Rank-One Updates
 $Q(i)$ is a rank-one update away from $QQ^\top$. There is no need for a $O(m^2N)$ recomputation per LOO.

3
Null Space Solve
 Find all $s_1, \dots, s_{m/2}$ at once by solving for a singular triangular solve.
 $R^\top S = D$

4
Final Complexity
 $O(m^3)$ inversion + $O(m^2N)$ block product. Cost is equal to Hutch++, therefore exchange ability is free.

XNysTrace Estimator

For $\mathbf{A} \in \mathbb{R}^{N \times N}$ positive semidefinite (PSD). Uses Nyström Approximation.

Nyström Approximation

$$\mathbf{A}\langle \mathbf{X} \rangle := \mathbf{A}\mathbf{X}(\mathbf{X}^* \mathbf{A}\mathbf{X})^\dagger (\mathbf{A}\mathbf{X})^*$$

Requires only k matvecs. Gram matrix allows each LOO update to reduce to a scalar with no recomputation

Key Difference

Utilizes all m vectors for each LOO approx, achieving a higher and more effective approx. rank $\approx m$ vs. $m/2$ for XTrace.

vs Hutch++

Converges roughly $3\times$ for matrices with exponentially decaying eigenvalues.

Limitation

Struggles with slowly decaying eigenvalues.

A Posterior Error Estimate

Error Estimate

Reuse the $\ell = m/2$ individual estimators already computed:

$$\widehat{\text{err}}^2 = \frac{1}{\ell(\ell-1)} \sum_{i=1}^{\ell} (\widehat{\text{tr}}_i - \widehat{\text{tr}}_X)^2$$

No extra cost

Derived from the $m/2$ estimators already computed, therefore zero additional matvecs.

Reliable Bounds

Proposition 3.1: ratio of error estimate to true squared error depends only on correlation between any two ℓ_i -s

Practical

Method is cheap and reliable due to a stopping mechanism that guarantees performance

Adaptive Stopping

Goal: meet relative tolerance $|\text{tr } \mathbf{A} - \widehat{\text{tr}}| \leq \varepsilon |\text{tr } \mathbf{A}|$ without fixing m in advance.

- ① Collect m_0 matvecs $\mathbf{A}\omega_1, \dots, \mathbf{A}\omega_{m_0}$, setting $j \leftarrow 0$.
- ② Form trace estimate $\widehat{\text{tr}}^{(j)}$ and error estimate $\widehat{\text{err}}^{(j)}$.
- ③ **If** $\widehat{\text{err}}^{(j)} \leq \varepsilon |\widehat{\text{tr}}^{(j)}|$: **stop**.
- ④ **Else**: collect m_j additional matvecs, set $j \leftarrow j + 1$, $m_j \leftarrow 2m_{j-1}$, then proceed with steps 2 through 4 until tolerance is met

Notes

- Uses at most $2\times$ the optimal number of matvecs
- each doubling adds as many products as all prior rounds, preserving $O(m^2 N)$ while converging in very few rounds
- Fast convergence of XTrace/XNysTrace results in very few doublings

Complete Variance Bounds

One can show the following:

Theorem 1.1. Let $A \in \mathbb{R}^{N \times N}$ be a square matrix. Fix the number of matvecs to be $m \geq 8$. All three of the Hutch++, XTrace, and XNysTrace estimators are unbiased estimators of the trace. With standard normal test vectors w , the estimators satisfy the variance bounds.

$$\begin{aligned}
 \left(\mathbb{E} |\widehat{\text{tr}}_{\text{H++}} - \text{tr } A|^2 \right)^{1/2} &\leq \min_{r \leq m/3-2} \frac{\sqrt{2} \|A - [A]_r\|_F}{\sqrt{m/3 - r - 1}}, \\
 \left(\mathbb{E} |\widehat{\text{tr}}_{\text{X}} - \text{tr } A|^2 \right)^{1/2} &\leq \sqrt{m} \min_{r \leq m/2-4} \left(\frac{2\|A - [A]_r\|}{\sqrt{m/2 - r - 3}} + \frac{2e\|A - [A]_r\|_F}{m/2 - r - 3} \right), \\
 \left(\mathbb{E} |\widehat{\text{tr}}_{\text{XN}} - \text{tr } A|^2 \right)^{1/2} &\leq m \min_{r \leq m-6} \left(\sqrt{8} \frac{\|A - [A]_r\|}{m - r - 5} + \sqrt{2} \frac{\|A - [A]_r\|_F}{(m - r - 5)^{3/2}} \right. \\
 &\quad \left. + \frac{\|A - [A]_r\|_*}{(m - r - 5)^2} \right).
 \end{aligned}$$

Complete Variance Bounds

Method	Frob. denominator	Exp. decay rate
Hutch++	$\sqrt{m/3 - r}$	$\alpha^{m/3}$
XTrace	$m/2 - r$	$\sqrt{m} \alpha^{m/2}$
XNysTrace	$(m - r)^{3/2}$	$m \alpha^m$

- XTrace converges $1.5\times$ faster than Hutch++
- XNysTrace converges $3\times$ faster than Hutch++
- All achieve $O(1/m^2)$ variance, superior to Hutch's $\Theta(1/m)$
- Nuclear norm term in XNysTrace bound penalises slow spectral decay

Proof Idea: Variance Bound

Goal: bound $(\mathbb{E} |\widehat{\text{tr}} - \text{tr } \mathbf{A}|^2)^{1/2}$ for each estimator.

- ① **Decompose variance into two terms.** By exchangeability of $\omega_1, \dots, \omega_{m/2}$:

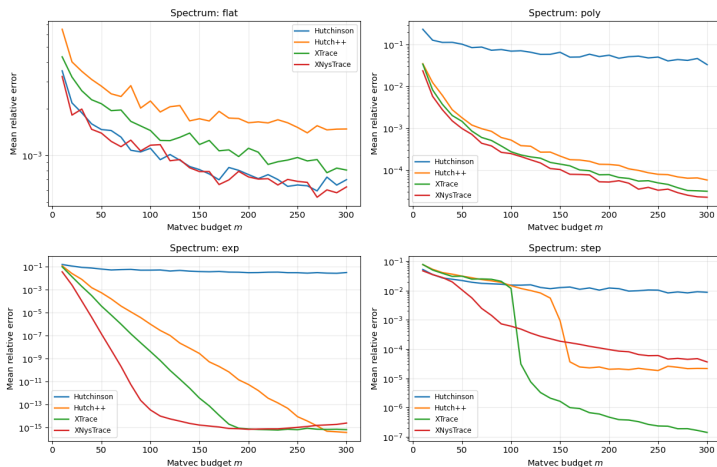
$$\text{Var}[\widehat{\text{tr}}] = \underbrace{\frac{1}{\ell} \text{Var}[\widehat{\text{tr}}_1]}_{\text{A: single estimator variance}} + \underbrace{\left(1 - \frac{1}{\ell}\right) \text{Cov}[\widehat{\text{tr}}_1, \widehat{\text{tr}}_2]}_{\text{B: correlation between estimators}}$$

where $\ell = m/2$ for XTrace and $\ell = m$ for XNysTrace.

- ② **Bound each term.**
- $\mathbf{A} \Rightarrow$ Frobenius norm of residual
 - $\mathbf{B} \Rightarrow$ Spectral norm of residual
- ③ **Substitute randomized SVD / Nyström error bounds**
- Replaces random quantities with deterministic $\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\|$.
 - Introduces nuclear norm for XNysTrace.
- ④ **Optimize over rank r .** Choosing $r \approx m/4$ (XTrace), $m/2$ (XNysTrace)

Reproduction of Spectra Simulation

Figure 3 reproduction — mean relative error vs. matvec budget
($N=1000$, 100 trials per point)



Extension: Simulated Networks & Trace Measures

Graph models ($N = 2000$)

Model	Structure & spectrum
ER ($c=3$, $c=10$)	Edges drawn i.i.d. with $p = c/N$.
SBM (20 blocks)	Dense edges within communities, sparse edges between community.
BA ($m=3$)	Generates hub nodes and a power-law degree distribution.

Trace measures

Length 4 closed walks

$$\text{tr}(M^4) = \sum_i \lambda_i^4$$

- Counts closed walks of length 4 in G
- Chosen because $M^4 \succeq 0$

Estrada index

$$\text{tr}(e^M) = \sum_i e^{\lambda_i}$$

- Measures graph “communicability” via all walk lengths
- Also has $e^M \succ 0$

Extension: Graph Trace Measures on Simulated Networks

Extension: trace of matrix functions on simulated graphs

